Solution: Problem Challenge 2: String Anagrams

Problem Statement

Given a string and a pattern, find **all anagrams of the pattern in the given string**.

Every **anagram** is a **permutation** of a string. As we know, when we are not allowed to repeat characters while finding permutations of a string, we get $N!$ permutations (or anagrams) of a string having N characters. For example, here are the six anagrams of the string “abc”:

1. abc
2. acb
3. bac
4. bca
5. cab
6. cba

Write a function to return a list of starting indices of the anagrams of the pattern in the given string.

Example 1:

```
Input: String="ppqp", Pattern="pq"
Output: [1, 2]
Explanation: The two anagrams of the pattern in the given string are "pq
" and "qp".
```

Example 2:

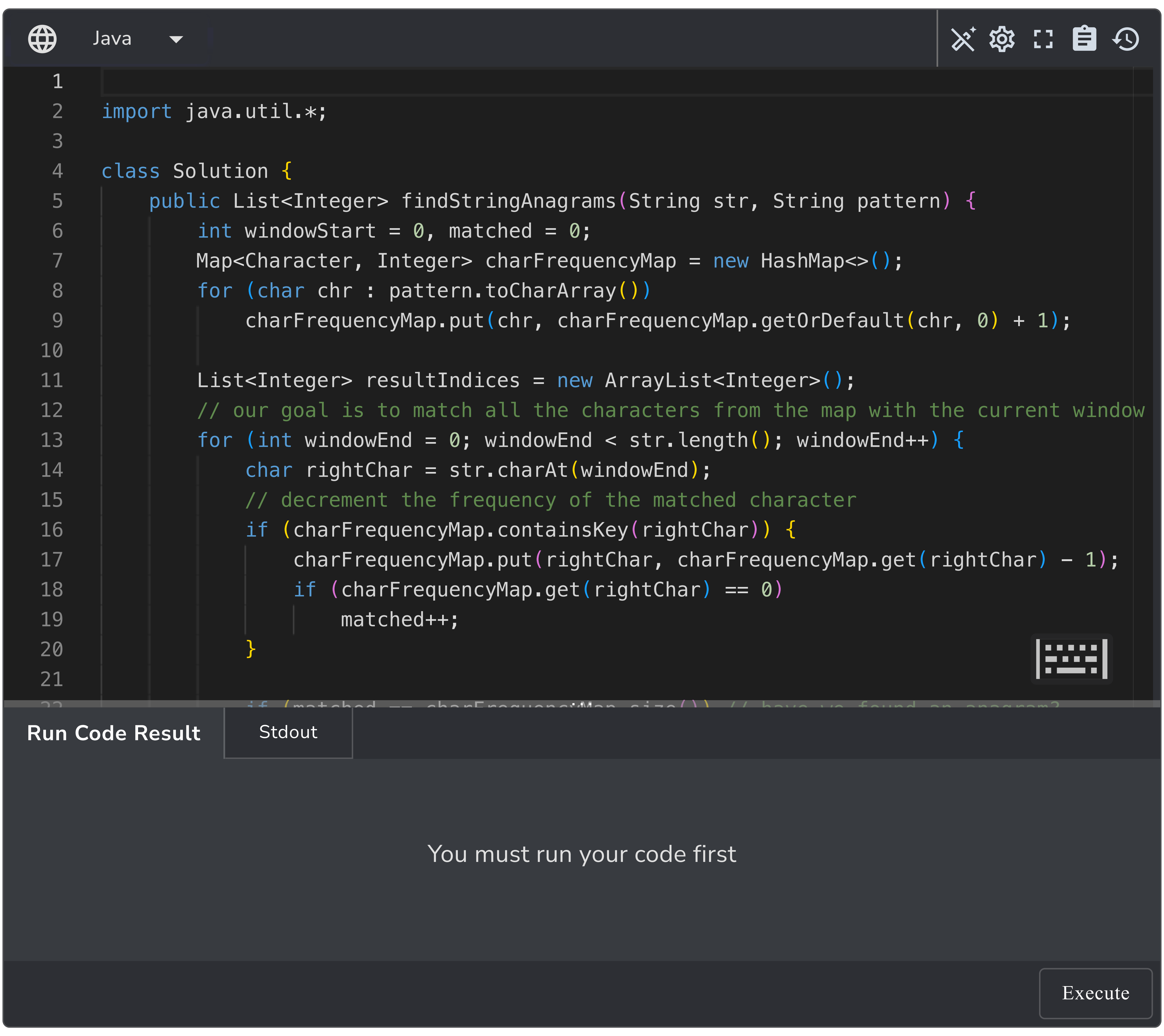
```
Input: String="abbcabc", Pattern="abc"
Output: [2, 3, 4]
Explanation: The three anagrams of the pattern in the given string are "
bca", "cab", and "abc".
```

Solution

This problem follows the **Sliding Window** pattern and is very similar to Permutation in a String. In this problem, we need to find every occurrence of any permutation of the pattern in the string. We will use a list to store the starting indices of the anagrams of the pattern in the string.

Code

Here is what our algorithm will look like:



```java

import java.util.*;

class Solution {
    public List<Integer> findStringAnagrams(String str, String pattern) {
        int windowStart = 0, matched = 0;
        Map<Character, Integer> charFrequencyMap = new HashMap<>();
        for (char chr : pattern.toCharArray())
            charFrequencyMap.put(chr, charFrequencyMap.getOrDefault(chr, 0) + 1);

        List<Integer> resultIndices = new ArrayList<Integer>();
        // our goal is to match all the characters from the map with the current window
        for (int windowEnd = 0; windowEnd < str.length(); windowEnd++) {
            char rightChar = str.charAt(windowEnd);
            // decrement the frequency of the matched character
            if (charFrequencyMap.containsKey(rightChar)) {
                charFrequencyMap.put(rightChar, charFrequencyMap.get(rightChar) - 1);
                if (charFrequencyMap.get(rightChar) == 0)
                    matched++;
            }

```

Time Complexity

The time complexity of the above algorithm will be $O(N+M)$ where ‘N’ and ‘M’ are the number of characters in the input string and the pattern respectively.

Space Complexity

The space complexity of the algorithm is $O(M)$ since in the worst case, the whole pattern can have distinct characters which will go into the **HashMap**. In the worst case, we also need $O(N)$ space for the result list, this will happen when the pattern has only one character and the string contains only that character.

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